

ASHWINI AHER

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WORK EXPERIENCE

Cellular System Test Engineer – Qualcomm, San Diego

July 2022-Dec 2023

- Executed 100+ 4G/5G test cases per release cycle to validate modem software using Qualcomm proprietary tools, covering SA/NSA call flow, uplink/downlink throughput, Carrier Aggregation (CA) and higher-order modulation schemes.
- Debugged test failures using QXDM, automation logs, and test setup analysis to identify root causes.
- Analyzed functional regressions, system crashes, and collaborated with development teams for timely resolution.
- Created and maintained Confluence documentation for new findings, knowledge sharing and training.

Modem Integration and Test Engineer – Qualcomm, San Diego (Contract - Novus Resources, Inc.)

Nov 2020-July 2022

- Validated modem software across 5G NSA/SA and 4G use cases, identifying 30% more regressions compared to team average.
- Collaborated with cross-functional teams to integrate, validate and support commercialization of Qualcomm 5G products.

Software Quality Analyst – Samsung Electronics America, Overland Park (Contract - BTI Solutions, Inc.)

July 2020-Nov 2020

- Evaluated SAMSUNG UE UI and modem software quality through field testing across 5G and LTE scenarios.
- Executed test cases (attach-detach, throughput, modulation) and collected logs using QXDM and on-device tools.

PROJECTS

Speech Based Channel Quality Classification (Python, PostgreSQL, Pandas, Scikit-learn, Librosa, VS Code)

April 2026

- Developed a Speech-Based Channel Quality Classification System using the Google Speech Commands dataset.
- Simulated channel conditions by adding AWGN noise at multiple SNR levels to generate realistic noisy speech signals.
- Extracted audio features such as MFCC, delta MFCC, and spectral features using Librosa.
- Performed data analysis and outlier detection in PostgreSQL, and created datasets with and without outliers.
- Trained Logistic Regression and SVM models using Scikit-learn, achieving ~92% classification accuracy on test data.

Student Performance Indicator Project (Python, Flask, Docker, AWS EC2, GitHub Actions, CI/CD)

December 2025

- Built an end-to-end ML pipeline for student performance prediction using Python, Pandas and Scikit-learn.
- Containerized and deployed a Flask-based application on AWS EC2 instance using Docker to enable reproducible predictions.
- Implemented a CI/CD pipeline using GitHub Actions to automatically build Docker images, push them to Amazon ECR, and deploy updated containers to EC2.

SINR Prediction Using Regression (Python, PostgreSQL, Pandas, Scikit-learn, VS Code)

March 2026

- Built an end-to-end ML pipeline integrating PostgreSQL with Python for data ingestion, preprocessing, and modeling.
- Simulated a wireless dataset using link-budget principles, including features as distance, path loss, received power, and SNR.
- Trained and evaluated regression models (Linear Regression, Ridge, Lasso) to predict SINR, achieving $R^2 \approx 0.93$ on test data.

Netflix Data Cleaning Data Analysis Project (PostgreSQL, Pandas)

January 2026

- Performed data cleaning and transformation on Netflix dataset using SQL (CTEs, window functions).
- Designed a normalized schema (ERD) and analyzed trends across director, genre, and country-wise content production.

EDUCATION

- **Master of Science in Electrical Engineering**, The State University of New York, USA

GPA: 3.74 | Sept 2019

- **Master of Engineering in Electronics and Telecommunication**, Mumbai University, India

GPA: 3.67 | Jan 2015

- **Bachelor of Technology in Electronics and Telecommunication**, Dr. B. A. T. U., India

GPA: 3.6 | June 2010

SKILLS

- **Programming:** Python, SQL
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Librosa, Matplotlib
- **Databases:** PostgreSQL, SQLAlchemy
- **Cloud & Deployment:** Docker, AWS EC2, Amazon ECR, Flask
- **DevOps:** Git, GitHub, GitHub Actions, CI/CD, PyTest
- **Tools:** VS Code, Jupyter Notebooks, QXDM

RESEARCH EXPERIENCE

Comparison of Vocal Tract Shape Estimation Based on Autocorrelation, Covariance, Lattice and Formant Frequencies. Dec 2014

- Utilized various Linear Predictive Coding (LPC) techniques to estimate Vocal Tract Shape using MATLAB.
- Published a research paper as a main author at ICNTE, India, (<http://ieeexplore.ieee.org/document/7029934/>).

CERTIFICATION

- Completed “Neural Networks and Deep Learning” course authorized by DeepLearning.AI, from Coursera.

March 2025